

CHURCHILL, MB, Canada

WMO: 716180

Lat: 58.740N

Lon: 94.070W

Elev: 96

StdP: 14.64

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.6	-31.6	-43.0	0.5	-34.3	-40.1	0.6	-31.6	35.4	-8.8	32.2	-9.7	11.8	290	0.960

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	78.3	63.3	73.8	60.8	69.5	58.8	64.9	74.9	62.3	71.0	59.8	67.8	12.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	80.8	69.4	58.3	72.8	66.7	55.7	66.2	64.0	30.1	75.0	28.2	71.1	26.3	67.6	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
32.5	28.9	25.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-38.2	86.2	4.0	2.6	-41.1	88.0	-43.4	89.6	-45.6	91.0	-48.5	92.9		
			-38.4	70.3	3.9	1.8	-41.2	71.6	-43.5	72.6	-45.7	73.6	-48.5	74.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.8	-13.5	-11.3	-1.3	14.8	30.4	46.1	55.7	54.6	45.2	31.6	11.2	-4.9
	DBStd	26.77	11.84	11.75	13.61	12.78	9.98	9.50	7.86	6.30	7.11	7.26	12.08	13.24
	HDD50	10793	1967	1715	1589	1055	613	187	32	20	179	571	1165	1700
	HDD65	15816	2432	2135	2054	1505	1073	571	305	332	594	1035	1615	2165
	CDD50	484	0	0	0	0	6	70	209	162	36	1	0	0
	CDD65	30	0	0	0	0	0	4	17	8	1	0	0	0
	CDH74	355	0	0	0	0	4	77	192	77	5	0	0	0
	CDH80	69	0	0	0	0	1	14	39	14	1	0	0	0
Wind	WSAvg	12.9	13.5	13.6	13.2	13.2	12.7	11.7	10.9	11.8	13.5	14.2	13.9	13.0
Precipitation	PrecAvg	17.7	0.8	0.6	0.8	1.0	1.5	1.9	2.2	2.6	2.4	2.0	1.5	0.9
	PrecMax	27.1	2.4	1.5	1.7	3.1	5.1	5.5	5.2	5.1	6.3	4.1	3.9	1.9
	PrecMin	12.2	0.2	0.1	0.1	0.0	0.2	0.4	0.4	0.6	0.8	0.5	0.4	0.0
	PrecStd	3.8	0.5	0.4	0.4	0.8	1.1	1.3	0.9	1.0	1.1	0.9	0.7	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.3	25.7	38.7	49.5	70.9	81.7	84.6	82.2	72.8	55.6	36.7	27.6
		MCWB	21.4	23.7	34.5	41.4	55.5	61.8	66.0	66.2	60.3	48.2	34.6	26.5
	2%	DB	15.7	16.9	32.4	42.0	58.3	75.2	79.6	75.2	65.9	47.2	32.8	23.7
		MCWB	15.0	15.6	29.3	36.2	48.7	59.3	64.3	62.7	57.1	43.3	31.8	22.5
	5%	DB	9.7	11.6	25.9	37.2	50.3	69.7	74.6	71.1	60.1	43.1	30.5	20.1
		MCWB	9.0	10.5	23.9	33.4	43.9	56.3	61.4	60.8	53.8	40.8	29.4	19.1
	10%	DB	4.3	7.1	19.4	33.0	43.7	63.4	70.0	66.1	55.6	40.2	28.0	15.8
		MCWB	3.6	6.1	17.9	30.6	39.3	54.1	59.7	58.5	51.2	38.5	27.0	15.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.5	24.0	35.0	42.1	56.5	65.1	69.0	68.3	62.0	48.7	34.9	26.8
		MCDB	22.1	25.3	38.7	48.6	69.1	77.2	80.7	79.0	70.0	55.4	36.5	27.4
	2%	WB	15.1	15.7	29.5	36.6	49.0	60.2	65.5	64.7	57.9	43.9	32.2	22.5
		MCDB	15.8	16.9	32.2	41.3	58.0	72.5	75.9	73.5	64.8	46.5	32.7	23.5
	5%	WB	9.0	10.6	23.9	33.7	44.1	57.2	63.0	61.7	54.3	41.2	29.5	19.2
		MCDB	9.6	11.5	25.9	36.8	50.1	68.9	72.2	69.1	58.9	42.9	30.2	20.1
	10%	WB	3.6	6.1	18.0	30.7	39.4	54.2	60.6	58.9	51.3	38.7	27.0	15.1
		MCDB	4.2	7.0	19.4	33.1	43.6	63.5	68.7	65.1	55.1	40.0	28.0	15.8
Mean Daily Temperature Range	5% DB	MDBR	13.2	14.3	16.6	16.1	12.7	17.6	17.3	14.0	11.4	8.3	12.3	13.0
		MCDBR	16.3	19.2	21.9	21.9	25.6	29.7	26.4	24.0	20.9	13.5	10.4	15.4
		MCWBR	16.2	18.5	19.7	17.2	17.4	16.6	13.6	13.1	13.2	10.3	10.1	15.1
	5% WB	MCDBR	16.3	19.1	21.7	20.6	24.8	28.4	23.9	21.9	19.3	12.5	10.2	15.2
		MCWBR	16.2	18.4	19.5	16.6	17.1	17.4	14.0	13.1	13.1	10.0	10.0	14.9
Clear-Sky Solar Irradiance	taub		0.208	0.251	0.278	0.317	0.336	0.339	0.361	0.339	0.310	0.290	0.232	0.185
	taud		2.126	2.159	2.155	2.091	2.173	2.344	2.442	2.524	2.578	2.486	2.211	2.046
	Ebn at Noon		207	249	274	280	283	284	275	274	267	238	192	174
	Edh at Noon		15	24	33	42	42	36	32	28	22	18	13	11
All-Sky Solar Radiation	RadAvg		186	509	1022	1581	1698	1900	1714	1289	792	360	161	109
	RadStd		15	34	82	147	185	170	111	102	77	37	23	9

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page